

TrustChain: Verified Informal Economy

Decentralized Soulbound Reputation & Credential Registry on Stellar & Soroban

Live Production:	https://trust-chain-mocha.vercel.app	Network:	Stellar Mainnet / Soroban Testnet
GitHub Repo:	https://github.com/OmcarSN/TrustChain.git	CI/CD Status:	Passing (8/8 green)
Tech Stack:	Stellar SDK, Soroban Rust, React 19, Supabase	Automated Tests:	131 Total (93 Frontend + 38 Rust)

1. Executive Summary & Problem Statement

Over **2 billion workers globally** operate in the informal economy (painters, carpenters, domestic staff, electricians, transport drivers). These workers face systematic financial and professional exclusion because their work history, skills, and client satisfaction remain locked in unverified word-of-mouth networks. Traditional platforms impose high fee commissions (15-30%) and custody worker identity on centralized servers.

TrustChain resolves this systemic problem by issuing non-transferable (soulbound) on-chain credentials and immutable peer reviews directly to Stellar wallets. Through an innovative fee-sponsorship engine, non-technical workers pay **zero gas fees** and require zero upfront cryptocurrency, bringing trust and verifiable history to informal labor markets.

2. Hybrid Architecture (Stellar Native + Soroban)

TrustChain operates a high-efficiency dual-layer blockchain design that balances zero-cost user accessibility with smart contract programmability:

Layer	Technology	Primary Responsibilities	Economic Benefit
Layer 1 (Native)	Stellar Classic ManageData + Horizon	High-frequency credential minting, endorsement records, key-value indexing, fee sponsorship top-ups.	Near-zero fee (~0.00001 XLM). Enables 100% sponsored gasless UX for workers.
Layer 2 (Smart Contracts)	Soroban VM (Rust Contracts)	Verification tier state machine, time-decay reputation scoring, right-of-reply on reviews, worker availability toggle, and DAO governance.	Cryptographic enforceability, tamper-proof state upgrades, multi-sig governance.

3. Core Product Features & Level 7 Capabilities

1. Gasless Worker Onboarding	Workers authenticate via Twilio phone OTP. The backend sponsor account funds their wallet with minimum balance reserve and signs transaction envelopes gaslessly.
2. Soulbound Credentialing	Credentials bind permanently to worker public keys, storing trade category, experience, phone metadata, and verification tier. Non-transferable by protocol design.
3. Anti-Sybil Endorsement Protocol	Employers sign weighted reviews (1-5 stars + text feedback). Real-time on-chain duplicate checks prevent spamming multiple reviews on the same worker profile.
4. Worker Availability Toggle	Interactive on/off switch on the profile page allowing workers to signal 'Available for Work' vs 'Unavailable'. Persisted and synchronized across discovery filters.
5. Right-of-Reply Mechanism	Workers can submit counter-responses to any client review directly via Soroban contract invocation, preventing one-sided defamatory reviews.
6. Multi-Filter Discovery & Explorer	Real-time public search by 18 trade categories, city, minimum rating threshold, availability status, and sorting by rating/review volume.
7. Full Hindi / English Localization	Bilingual i18n interface designed specifically for informal workers across India and developing economies, switchable with one click in the header.

4. Smart Contract Registry & Verification

Contract	Network	Contract Address (Soroban)	Status
Credential	Mainnet	CCFRAOH3L2Q7N4OWXG2GGUHL7NH4RD4XX57VZLCJ37P4YTAMNDAK5T63	Verified
Reputation	Mainnet	CBEIPVBJDHGCCTIPLR6XEVK6LQZVMTCC6PKUP5Y5KKVUZRL50E7L7QDW	Verified
Credential	Testnet	CCQG7ZF0QX4H70SUXDU2FE2J73XCUQLXP2F0NFUMKXXDUP253J3JP6HZ	Active
Reputation	Testnet	CAHH72L2C7XN32IK0IBLN0DWQF4MLUJTJRZWFCU4JIUQEXLEEX3E52GI	Active
Governance	Mainnet/Testnet	Source in contracts/governance/src/lib.rs (DAO UI at /governance)	Built (Ready)

5. Security Hardening & CI/CD Pipeline

TrustChain implements comprehensive automated verification ensuring zero downtime and audited smart contract execution:

Stage / Check	Tool / Target	Coverage / Specification	Result
Frontend Unit Tests	Vitest + React Testing Library	93 tests covering UI components, accessibility, fee bump, and forms	93/93 Passed
Smart Contract Tests	Cargo test (Soroban Rust SDK)	38 contract tests covering state machines, auth checks, and arithmetic	38/38 Passed
Rust Clippy & Linter	Cargo clippy --all-targets	Enforces zero compiler warnings and strict code hygiene	0 Warnings
Security & Rate Limiting	Vercel Serverless + SECURITY.md	5 req/min/IP sliding window, CORS protection, XSS sanitized inputs	Active
Automated CI/CD	GitHub Actions (ci.yml)	8 parallel jobs validating Node 22 build, WASM optimize, and deploy	Passing

6. Roadmap & Next Milestones (Level 7 Founder Track)

- **DeFi Micro-Lending Pool:** Enabling informal workers to leverage their on-chain reputation score as under-collateralized loan credit.
- **Soroban Mainnet Governance Deployment:** Transitioning admin key management to the on-chain DAO multi-sig council.
- **Zero-Knowledge Verification (zk-SNARKs):** Enabling workers to prove rating and experience thresholds without revealing phone numbers or identity.
- **WhatsApp Bot Integration:** Allowing offline and low-bandwidth workers to manage availability and credentials via conversational AI.

Official Links & References:

- **Live Application:** <https://trust-chain-mocha.vercel.app>
- **GitHub Repository:** <https://github.com/OmcarSN/TrustChain.git>
- **Official Twitter/X:** <https://x.com/TrustChainXLM>
- **Stellar Expert Explorer:** <https://stellar.expert/explorer/public/contract/CCFRAOH3L2Q7N4OWXG2GGUHL7NH4RD4XX57VZLCJ37P4YTAMNDAK5T63>